

THE UNIVERSITY OF DODOMA



COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION

DEGREE PROGRAMME: BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

COURSE NAME: MICROCONTROLLER SYSTEMS.

COURSE CODE: CT 321.

GROUP NUMBER: 6.

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TWI/I²C Pin Mapping Table

ATmega32 Pin	Port Function	Configured As	Connected Peripheral Device	Purpose
PC0	SCL (Serial Clock)	Output (Open- Drain)	RTC SCL Pin	Generates the <i>I²C</i> clock signal.
PC1	SDA (Serial Data)	Bidirectional	RTC SDA Pin	Transfers address and time data bytes.
PA0	GPIO (Digital In)	Input (Internal Pull-Up)	"Set" Push Button	Enters time-setting configuration mode.
PA1	GPIO (Digital In)	Input (Internal Pull-Up)	"Increment" Push Button	Increments time values during setup
PA2	GPIO (Digital In)	Input (Internal Pull-Up)	"Confirm" Push Button	Saves the newly targeted time value
PB0	GPIO (Digital Out)	Output	Red LED	Illuminates if an <i>I²C</i> bus error occurs.
PD0	RXD (Optional)	Input	Serial Terminal RX	Receives debugging commands (if used).
PD1	TXD (Optional)	Output	Serial Terminal TX	Transmits time/date text to a PC terminal.

Basic RTC Register Map

Register Address	Register Name	Data Format	Range (BCD)	Description
0x00	Seconds	BCD	00 – 59	Holds current seconds. Bit 7 sometimes acts as the Clock Halt (CH) bit.
0x01	Minutes	BCD	00 – 59	Holds current minutes
0x02	Hours	BCD	00 – 23 or 01 – 12	Holds hours. Bit 6 configures 12-hour (AM/PM) or 24-hour mode.
0x03	Day of Week	BCD	01 – 07	Day of the week (e.g., 1 = Sunday, 2 = Monday)
0x04	Date (Day of Month)	BCD	01 – 31	Holds the calendar day of the month.
0x05	Month	BCD	01 – 12	Holds the calendar month
0x06	Year	BCD	00 – 99	Holds the two-digit year tracking (e.g., 26 for 2026).
0x07	Control / Status	Mixed	---	Configures square-wave output, oscillators, or alarm flags.